accuracy rather than being riddled with a variety of huge warps and severe curves, also like those generated by black holes?

As first discussed in detail by Paul Davies and Don Page,[3] inflationary cosmology gives important insight into these issues. To see how, bear in mind that an essential assumption of the puzzle is that once a clump forms here or there, its greater gravitational pull attracts yet more material, causing it to grow larger; correspondingly, once a wrinkle in space forms here or there, its greater gravitational pull tends to make the wrinkle yet more severe, leading to a bumpy, highly nonuniform spatial curvature. When gravity matters, ordinary, unremarkable, high-entropy configurations are lumpy and bumpy.

But note the following: this reasoning relies completely on the *attractive*nature of ordinary gravity. Lumps and bumps grow because they *pull* strongly on nearby material, coaxing such material to join the lump. During the brief inflationary phase, though, gravity was *repulsive* and that changed everything. Take the shape of space. The enormous outward push of repulsive gravity drove space to swell so swiftly that initial bumps and warps were stretched smooth, much as fully inflating a shriveled balloon stretches out its creased surface.[29] What's more, since the volume of space increased by a colossal factor during the brief inflationary period, the density of any clumps of matter was completely diluted, much as the density of fish in your aquarium would be diluted if the tank's volume suddenly increased to that of an Olympic swimming pool. Thus, although attractive gravity causes clumps of matter and creases of space to grow, repulsive gravity does the opposite: it causes them to diminish, leading to an ever smoother, ever more uniform outcome.

Thus, by the end of the inflationary burst, the size of the universe had grown fantastically, any nonuniformity in the curvature of space had been stretched away, and any initial clumps of anything at all had been diluted to the point of irrelevance. Moreover, as the inflaton field slid down to the bottom of its potential-energy bowl, bringing the burst of inflationary expansion to a close, it converted its pent-up energy into a nearly uniform bath of particles of ordinary matter throughout space (uniform up to the tiny but critical inhomogeneities coming from quantum jitters). In total, this all sounds like great progress. The outcome we've reached via